

6(a)	work done per unit charge	B1
	(work done in) moving positive charge from infinity	B1
6(b)	$C = Q / V$	C1
	$V = Q / (4\pi\epsilon_0 r)$ and so $C = Q / [Q / (4\pi\epsilon_0 r)] = 4\pi\epsilon_0 r$	A1
6(c)	$Q = 4\pi\epsilon_0 r V = 4\pi \times 8.85 \times 10^{-12} \times 0.13 \times 4500$ $(= 6.5 \times 10^{-8} \text{ C})$	C1
	$(Q - q) / 13 = q / 5.2$	C1
	$5.2Q - 5.2q = 13q$, so $q = (5.2 / 18.2)Q$ $q = (5.2 / 18.2) \times 6.5 \times 10^{-8}$ $= 1.9 \times 10^{-8} \text{ C}$	A1
	or	
	$V_T = Q_T / C_T$ $= 6.5 \times 10^{-8} / [4\pi \times 8.85 \times 10^{-12} \times (0.13 + 0.052)]$ $(= 3210 \text{ V})$	(C1)
	$q = 4\pi \times 8.85 \times 10^{-12} \times 0.052 \times 3210$ $= 1.9 \times 10^{-8} \text{ C}$	(A1)